

A Rare Presentation Of IgG Multiple Myeloma

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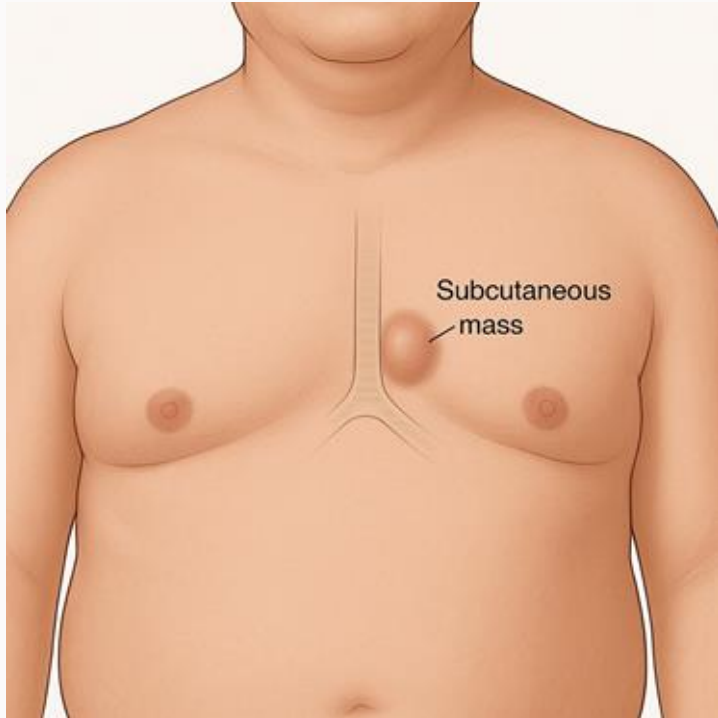


Disclosures

All presenters and authors have nothing to disclose.

*All presenters/authors **attest** that clinical recommendations are **evidence-based** and **free of commercial bias***

“Chest pain and a lump on my chest!”



68yo M

3-month history of chest pain associated with an enlarging left parasternal mass

HPI - 68yo M with a PMHx:

IDT2DM (last HbA1c 7.2) (on Insulin Glargine 160U QD, Dapagliflozin 10mg QD, and Glipizide 5mg QD) c/b **CKD Stage 3a**, **CAD** s/p DES x 3 c/b angina (last DES to OM on 08/2019) (on Ticagrelor 60mg QD, Aspirin 81mg QD, Carvedilol 25mg BID and SL Nitroglycerin 0.4mg PRN), **HLD** (on Atorvastatin 80mg QD), **HTN** (on Hydralazine 50mg BID, HCTZ 25mg QD), **Hypothyroidism** (on Levothyroxine 225mcg acbr) **HFpEF** (on Bumetanide 1mg QD and Potassium Chloride 20mEq QD)

COPD (on Albuterol PRN)
GERD (on Omeprazole 20mg QD)
Erectile dysfunction (on Tadalafil 20mg PRN)
Glaucoma (on Latanoprost 1 drop b/l QD), **Chronic left knee pain** (on Celecoxib 200mg PRN),
Obesity (BMI 30)
Chronic hepatitis C.
PSH: None
SH: Denies any tobacco, alcohol or illicit drug use
FH: None
Allergies: None

Physical Exam

Vital Signs: Stable (T 36.5, Pulse 67, RR 18, **BP 160/101**, SpO2 100%)

General: Awake, alert, cooperative, in no acute distress

Cardiac: RRR, no murmurs

Neck: Trachea midline, no JVD

Chest Wall: **Palpable firm mass at the left parasternal chest wall, mildly tender, pain is worse with arm elevation above his head**

Lungs: CTA

Abdominal: soft, non-distended, non-tender, normoactive bowel sounds

Neuro: Alert and oriented x3, moving all extremities spontaneously, no focal deficits

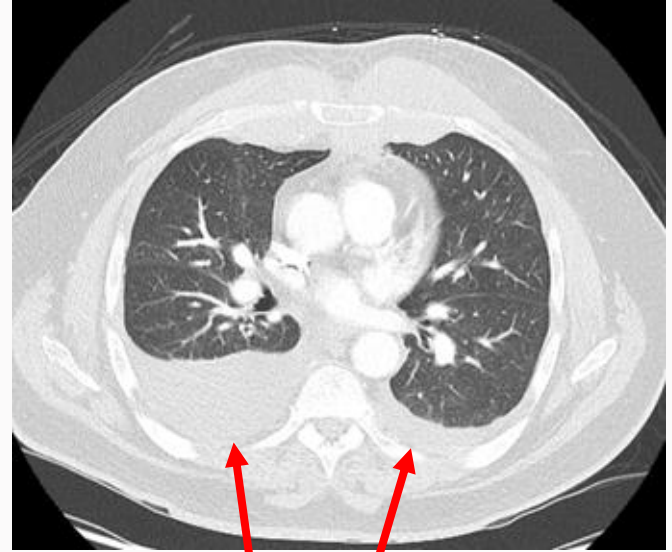
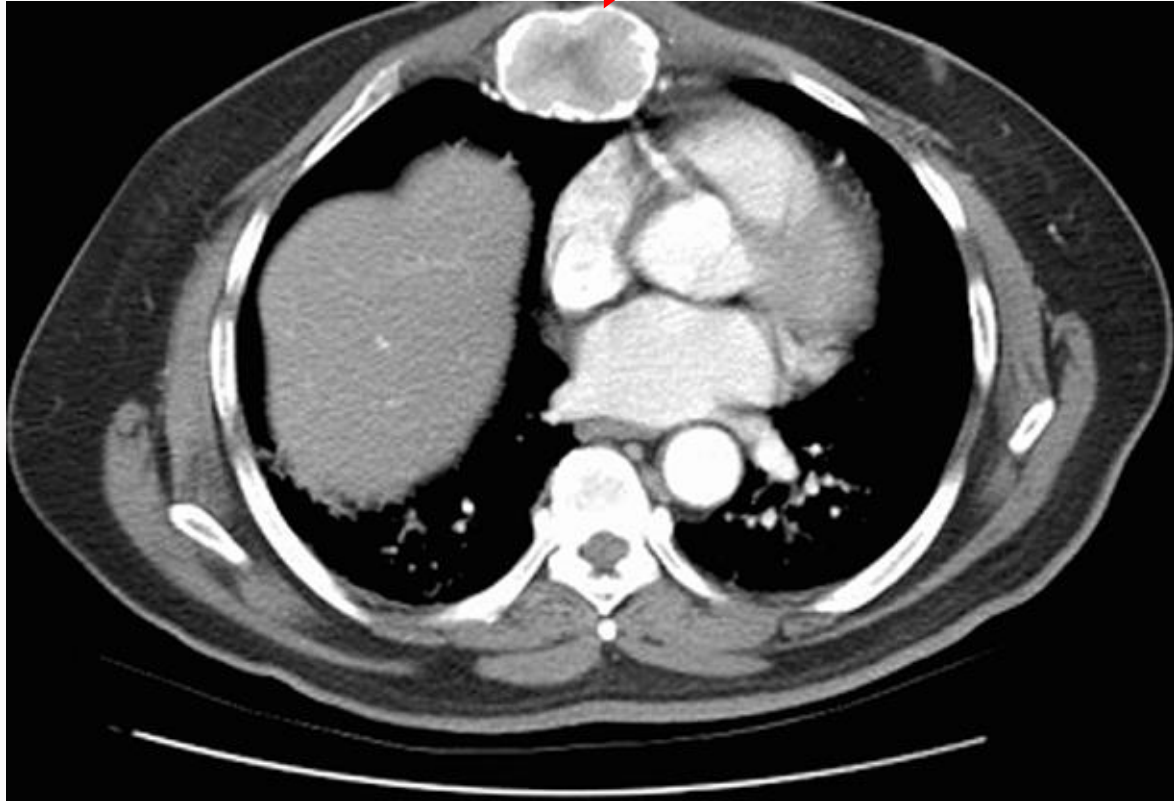
Skin: Warm, dry, intact

Notable Labs

High**Low**

CBC	CMP	Miscellaneous	
WBC 4.67K	Glucose 194	AFP < 10	Uric Acid 7.8
RBC 3.59	Cr 1.33	β-hCG < 1	Trop 75 (0hr), 80 (1hr), 75(6hr)
Hgb 11.0	Na 132	LDH 353	INR 1.3
Plt 201K	Ca 8.6	Iron Level 78	Total Protein 8.3
MCV 94	K 4.4	Ferritin 148	Transferrin 164
CRP 71	HCO3 26	Albumin 3.51	TSH 2.68

Imaging



CT Chest W

Neoplastic Differential

Anterior Mediastinal Mass

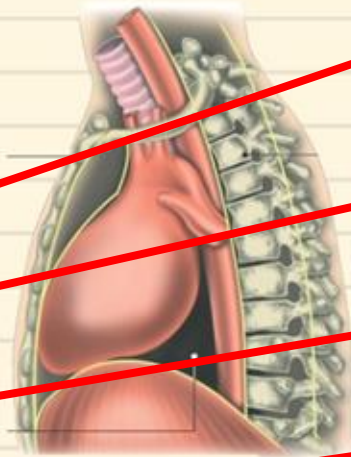
4Ts

Thymoma

Teratoma

Thyroid goiter

"Terrible" Lymphoma



AChR & MuSK Ab (to screen for MG) (present in about 40% of thymoma patients)

AFP & Beta-hCG & Testicular USN

TSH & Free T4 & Thyroglobulin

LDH +/- Uric Acid

A 5th T?

Transfers aka Metastases

Not All Mediastinal Masses Are Primary

Possible Metastatic Sources

- Lung carcinoma – most common intrathoracic primary
- Breast carcinoma – can spread to anterior mediastinum and sternum
- Renal cell carcinoma – hematogenous spread; can produce hypervascular lesions
- Thyroid carcinoma (follicular) – can metastasize to bone or mediastinum
- Melanoma



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Hyperviscosity Syndrome

Causes		
Increased Immunoglobulins	Increased Blood Cells	Rare: Rigid RBCs
Multiple Myeloma (usually IgA Myeloma) Waldenstrom's Macroglobulinemia CLL	Leukemias Polycythemia Vera Essential Thrombocythemia Primary Myelofibrosis	Sickle Cell Disease Hereditary Spherocytosis

Viscosity, Serum

* (H) 2.4

Tx: Plasmapheresis

Drug (Abbreviation)	Mechanism of Action (Simplified)	Clinical Pearl	
Daratumumab (Dara)	Monoclonal antibody targeting CD38 on plasma cells → triggers immune-mediated destruction (complement, NK-cell, and macrophage killing).	The first FDA-approved antibody for myeloma; can cause transient anemia or low counts during initial infusions because normal plasma cells also express CD38.	4-6 months of Dara-RVD ↓ Autologous Stem Cell Transplant ↓ Maintenance (Lenalidomide)
Lenalidomide (Revlimid, R)	Immunomodulatory agent (IMiD) enhances T-cell and NK-cell activity, suppresses pro-tumor cytokines, and induces myeloma cell apoptosis.	Derived from <i>thalidomide</i> (same parent compound) but far more potent and better tolerated; it's also used as maintenance therapy after transplant.	
Bortezomib (Velcade, V)	Proteasome inhibitor blocks degradation of misfolded proteins, leading to myeloma cell stress and apoptosis.	Revolutionized myeloma care in the early 2000s; given subcutaneously to reduce risk of peripheral neuropathy.	
Dexamethasone (d)	Corticosteroid induces apoptosis of lymphoid and plasma cells, reduces inflammation, and potentiates activity of the other agents.	Often the “most immediately effective” component; patients may feel dramatic improvement in pain or fatigue within days of starting therapy.	